Michio Honda

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INTERESTS

Computer Networks, Operating Systems and Security

EDUCATION

Keio University, Tokyo, Japan

Ph.D. in Graduate School of Media and Governance, Cyber Informatics — Apr 2009 – Mar 2012
Thesis: The Internet is not an Internet—Principles, Evasion and Implications for Transport Protocols

M.S. in Graduate School of Media and Governance, Cyber Informatics — Apr 2007 – Mar 2009
Thesis: Bidimensional-Probe Multipath Congestion Control for Shared Bottleneck Fairness

B.S. in Faculty of Environment and Information Studies — Apr 2003 – Mar 2007
Thesis: Fast Transport Layer Handover Using Single Wireless Interface

EMPLOYMENT

University of Edinburgh, School of Informatics, Edinburgh, UK
Reader (Associate Professor) — Aug 2024 – present
Lecturer in Networked Systems (Assistant Professor) — Jan 2020 – Jul 2024

Senior researcher, **NEC** Laboratories Europe, Heidelberg, Germany — Nov 2016 – Dec 2019
Software engineer, **NetApp**, Munich, Germany — Dec 2014 – Oct 2016
Research scientist, **NEC** Laboratories Europe, Heidelberg, Germany — Jul 2012 – Nov 2014

OTHER EXPERIENCE

Visiting student researcher, University College London (UCL), London, UK — Apr 2010 – Sep 2010
Advisor: Prof. Mark Handley
Focus: Middlebox, Multipath Transport Protocol.

Research intern, Nokia Research Center, Espoo, Finland — Jul 2008 – Jan 2009
Advisor: Dr. Lars Eggert
Focus: Multipath Transport Protocol.

AWARDS

Google Research Scholar Award * — Apr 2022
Facebook Research Award * — Aug 2021
Best paper award, ACM SOSR'15 * — Jun 2015
Community award, USENIX NSDI'12 — Apr 2012
IETF/IRTF Applied Networking Research Prize (ANRP) — Nov 2011* and Jan 2026
* as an individual recipient or lead author

SELECTED PROJECTS

Secure datacenter transport protocol — Oakland'26, NDSS'26
SMT is a new encrypted transport protocol designed for low-latency datacenter transports. It refines TLS with per-message record sequence number spaces to avoid head-of-line blocking, while preventing replay attacks. Looma enables low-latency TLS handshake with post-quantum resistance and mutual authentication in the cloud, using an online/offline signature scheme.

Host stack support for traffic analysis attack defenses — HotNets'25
Traffic analysis attacks like website fingerprinting is a growing threat to the encrypted Internet traffic. Stob is a framework that integrates traffic obfuscation directly in the host network stack. Its design rational is that packet-level features that the attacks exploit, like sizes, bursts and timing, are determined at the transport layer or below.

Flexible TCP load balancing — NSDI'26, NSDI'21
Prism is a new layer 7 load balancing (L7LB) architecture that enables the backend server to return the response to the client directly, bypasssing the L7LB device. Prism uses TCP connection migration with the aid of programmable switches; XO enables similar features to Prism but without switch support, and integration with real applications like Nginx and Ceph.

Network stack design for modern hardware — ATC'25, HotNets'21, NSDI'18, HotNets'16, ATC'16
PASTE is a network stack that offers unified abstractions of network and non-volatile main memory. It fills the gap between the storage and network stacks designed in isolation, and solves the problem with the costs of moving and transforming data between these stacks that are significant for non-volatile main memory that offers fast, byte-addressable persistence.

Scalable virtual networking — SOSR'15, CCR'14, NSDI'14, SoCC'17
mSwitch solves the scalability problem of existing software switches that is crucial to consolidate a large number of VMs or virtualized network functions by a novel packet forwarding algorithm and streamlined data path. It was initially designed for ClickOS, a tiny unikernel that runs Click, and MultiStack, a framework that runs multiple user-space network stacks.

Middlebox measurement for TCP extensibility — IMC'11, NSDI'12

This work was motivated by exploring viable design of Multipath TCP. It transmits various non-existent TCP traffic that mimics possible future TCP extensions to our server and examines on-path actions to the packets. This is the first work that examines in-depth middlebox behaviour prevalent in the Internet.

PROFESSIONAL SERVICE

UNIVERSITY OF EDINBURGH

People and Culture committee — 2022–present
Programming Club, organizer — 2020–2024

PROFESSIONAL SOCIETY

ACM SIGOPS, CARES Committee co-chair — 2023–present
IRTF ANRP Award Committee — 2022–present
ACM SIGOPS, CARES Committee — 2021–2022

CONFERENCE PROGRAM COMMITTEE

USENIX OSDI (2025–2027), NSDI (2023, 2026), ACM EuroSys (2021, 2024–2026), ACM SIGCOMM (2024–2026), ACM APSYS (2024–2025), ACM HotNets (2024), ACM ASPLOS (2024), USENIX/ACM ATC (2017, 2018, 2020–2024, 2026), ACM HotStorage (2022–2025), ACM/IEEE ANCS (2018, 2021 (co-chair)), ACM CoNEXT (2021–2022), ACM/IEEE SC (2019), ACM SOSR (2018), ACM EuroDW (2018, ACM/IEEE ToN (2017–2018), ACM SOSP poster (2013)

TEACHING

Introduction to Programming (Informatics) Summer, University of Edinburgh — Jul 2021 and 2022
Computer Communications and Networks, University of Edinburgh — Fall 2021–present
Data Structures and Programming, Keio University — Fall 2011
Fundamentals of Information Technology, Keio University — Spring 2011

MENTORING

UNIVERSITY OF EDINBURGH

Shuo Li, PhD supervision (graduated) — Nov 2021–Jul 2026
Xinshu Ma, PhD supervision — Jul 2023–present
Tianyi Gao, PhD supervision — Dec 2022–present
Elisaveta Lavrentieva, PhD supervision (equal co-supervision with Marc Juarez) — Sep 2024–present
Michael Zhang, Internship mentoring (ICSA Summer Internship) — Summer 2024
Eugenio Luo, Internship mentoring (EPSRC Vacation Internship) — Summer 2024
Tianyi Gao, MSc and intern supervision — Spring 2022
Steven W. D. Chien, Postdoc → assistant professor at the University of St Andrews. — Mar 2022–Sep 2025
Shinichi Awamoto, PhD supervision — Jan 2021–Jan 2024

NETAPP

Nanako Momiyama, BSc thesis supervision, Keio University — Fall 2016
Yutaro Hayakawa, BSc thesis intern at NetApp, Keio University — Fall 2016
Kenichi Yasukata, MSc thesis intern at NetApp, Keio University — Fall 2015

NEC LABS EUROPE

Shinichi Awamoto, intern at NEC, Tokyo University — Fall 2019
Yutaro Hayakawa, MSc thesis intern at NEC, Keio University — Fall 2018
Nanako Momiyama, intern at NEC — Spring 2017
Kenichi Yasukata, mentor at NEC — Fall 2016

GRANTS

GLOBAL

NetApp Faculty Fellowship, $50K, sole PI — Jan 2023
Towards Generic, Encrypted Datacenter Transport
Google Research Scholar Award, $60K, sole PI — May 2022
Upcycling Packets as Persistent In-Memory Data Structures
Facebook Research Award, $50K, sole PI — Nov 2021
Flexible transport scale-out with modern NICs

REGIONAL (UK/EU)

EPSRC Core Equipment Award, £50K (my share), Co-I — Jan 2025 – Jun 2026
(my part) Systems Research Testbed
Royal Society Research Grant, £20K, sole PI — Oct 2024 – Oct 2025
Confidential Computing at a Scale
EPSRC Core Equipment Award, £35K (my share), Co-I — Jan 2023 – Mar 2023
(my part) Systems Research Testbed
EPSRC New Investigator Award, £385K, sole PI — Apr 2022 – Mar 2025
NetPM: Co-designing Data Management and Networking Principles for Persistent Memory
NCSC RISE Proof-of-Concept, £45K, co-PI — Oct 2021 – Mar 2022
Follow on Project: Gupt - A Hardware Assisted Secure and Private Data Analytics

OTHER HONORS **Nominee for Teaching Award**, Outstanding Course category, University of Edinburgh — Apr 2023

STUDENT SCHOLARSHIP

Research Fellowship for Young Scientists (DC1) — Apr 2009 – Mar 2012
Japan Society for the Promotion of Science, 9.2M JPY

Excellent Young Researcher Overseas Visit Program — Apr 2010
Japan Society for the Promotion of Science, 1M JPY

Young Leader Scholarship — Apr 2009
Keio University, 1M JPY

SELECTED PUBLICATIONS

* denotes equal contribution or supervision role.

Tianyi Gao, Xinshu Ma, Suhas Narreddy, Eugenio Luo, Steven W. D. Chien and **Michio Honda**, *"Designing Transport-Level Encryption for Datacenter Networks"*, IEEE Symposium on Security and Privacy (**S&P "Oakland"**), May 2026. **IETF/IRTF Applied Networking Research Prize**

Shuo Li*, Steven W. D. Chien*, Tianyi Gao and **Michio Honda**, *"Remote TCP Connection Offload and Applications"*, USENIX Symposium on Networked Systems Design and Implementation (**NSDI**), May 2026.

Xinshu Ma and **Michio Honda**, *"Looma: A Low-Latency PQTLS Authentication Architecture for Cloud Applications"*, ISOC Network and Distributed System Security Symposium (**NDSS**), Feb 2026.

Elisaveta Lavrentieva, Marc Juarez* and **Michio Honda***, *"Rethinking the Role of Network Stacks for Website Fingerprinting Defenses"*, ACM Workshop on Hot Topics in Networks (**HotNets**), Nov 2025.

Steven W. D. Chien, Kento Sato, Artur Podobas, Niclas Jansson, Stefano Markidis and **Michio Honda**, *"ParaLog: Consistent Host-Side Logging for Parallel Checkpoints'*, ACM Symposium on Cloud Computing (**SoCC**), Nov 2025.

Shinichi Awamoto and **Michio Honda**, *"Opening Up Kernel-Bypass TCP Stacks"*, USENIX Annual Technical Conference (**ATC**), Jul 2025.

Michio Honda, *"Packets as Persistent In-Memory Data Structures"*, ACM Workshop on Hot Topics in Networks (**HotNets**), Nov 2021.

Yutaro Hayakawa, **Michio Honda**, Douglas Santry and Lars Eggert, *"Prism: Proxies without the Pain"*, USENIX Symposium on Networked Systems Design and Implementation (**NSDI**), Apr 2021.

Shinichi Awamoto, Erich Focht and **Michio Honda**, *"Designing a Storage Software Stack for Accelerators"*, USENIX Workshop on Hot Topics in Storage and File Systems (**HotStorage**), Jul 2020.

Maurice Bailleu, Jörg Thalheim, Pramod Bhatotia, Christof Fetzer, **Michio Honda** and Kapil Vaswani, *"Speicher:Securing LSM-based Key-Value Stores using Shielded Execution"*, USENIX Conference on File and Storage Technologies (**FAST**), Feb 2019.

Salvatore Pontarelli, Roberto Bifulco, Marco Bonola, Carmelo Cascone, Marco Spaziani, Valerio Bruschi, Davide Sanvito, Giuseppe Siracusano, Antonio Capone, **Michio Honda**, Felipe Huici and Giuseppe Bianchi, *"FlowBlaze: Stateful Packet Processing in Hardware"*, USENIX Symposium on Networked Systems Design and Implementation (**NSDI**), Feb 2019.

Michio Honda, Giuseppe Lettieri, Lars Eggert and Douglas Santry, *"PASTE: A Network Programming Interface for Non-Volatile Main Memory"*, USENIX Symposium on Networked Systems Design and Implementation (**NSDI**), Apr 2018.

Kenichi Yasukata, Felipe Huici, Vincenzo Maffione, Giuseppe Lettieri and **Michio Honda**, *"HyperNF: Building a High Performance, High Utilization and Fair NFV Platform"*, ACM Symposium on Cloud Computing (**SoCC**), Sep 2017.

Simon Kuenzer, Anton Ivanov, Filipe, Manco, Jose Mendes, Yuri Volchkov, Florian Schmidt, Kenichi Yasukata, **Michio Honda** and Felipe Huici, *"Unikernels Everywhere: The Case for Elastic CDNs"*, ACM International Conference on Virtual Execution Environments (**VEE**), Apr 2017.

Michio Honda, Lars Eggert and Douglas Santry, *"PASTE: Network Stacks Must Integrate with NVMM Abstractions"*, ACM Workshop on Hot Topics in Networks (**HotNets**), Nov 2016.

Kenichi Yasukata, **Michio Honda**, Douglas Santry and Lars Eggert, *"StackMap: Low-Latency Networking with the OS Stack and Dedicated NICs"*, USENIX Annual Technical Conference (**ATC**), Jun 2016.

Michio Honda, Felipe Huici, Giuseppe Lettieri and Luigi Rizzo, *"mSwitch: A Highly-Scalable, Modular Software Switch"*, ACM SIGCOMM Symposium on SDN Research (**SOSR**), Jun 2015. **Best paper award**

Michio Honda, Felipe Huici, Costin Raiciu, Joao Araujo and Luigi Rizzo, *"Rekindling Network Protocol Innovation with User-Level Stacks"*, ACM SIGCOMM Computer Communication Review (**CCR**), Apr 2014.

Joao Martins, Mohamed Ahmed, Costin Raiciu, Vladimir Olteanu, **Michio Honda**, Roberto Bifulco and Felipe Huici, *"ClickOS and the Art of Network Function Virtualization"*, USENIX Symposium on Networked Systems Design and Implementation (**NSDI**), Apr 2014.

Costin Raiciu, Christoph Paasch, Sebastien Barre, Alan Ford, **Michio Honda**, Fabien Duchene, Olivier Bonaventure and Mark Handley, *"How Hard Can It Be? Designing and Implementing a Deployable Multipath TCP"*, USENIX Symposium on Networked Systems Design and Implementation (**NSDI**), Apr 2012. **Community Award**

Michio Honda, Yoshifumi Nishida, Costin Raiciu, Adam Greenhalgh, Mark Handley and Hideyuki Tokuda, *"Is it Still Possible to Extend TCP?"* ACM Internet Measurement Conference (**IMC**), Nov 2011. **IETF/IRTF Applied Networking Research Prize**

Michio Honda, Yoshifumi Nishida, Pasi Sarolahti and Lars Eggert, *"Multipath Congestion Control for Shared Bottleneck"* International Workshop on Protocols for Future, Large-Scale & Diverse Network Transports (**PFLDNeT**), May 2008.

Michio Honda, Jin Nakazawa, Yoshifumi Nishida, Masahiro Kozuka and Hideyuki Tokuda, *"A Connectivity-Driven Retransmission Scheme Based On Transport Layer Readdressing"*, IEEE International Conference on Distributed Computing Systems (**ICDCS**), Jun 2008.

INTERNET DRAFTS

Xinshu Ma, Michio Honda and Colin Perkins, *"Looma: Low-Latency Post-Quantum Authentication for TLS 1.3 in Datacenters"*, draft-ma-cfrg-looma-00, Mar 2026.

OPEN SOURCE CONTRIBUTION
(by myself, not mentees)

PASTE	https://micchie.github.io/paste/
netmap	mSwitch and various features https://github.com/luigirizzo/netmap
MultiStack	https://github.com/sysml/multistack
Linux kernel	SCTP extensions https://www.kernel.org/
FreeBSD kernel	mSwitch and SCTP extensions https://www.freebsd.org/

[*CV compiled on 2026-07-21*]